Supplementary Material

The Injection Site Determines What Identity Survives in Training-Free Multi-Concept Personalization

This document holds results that support the manuscript without carrying its argument: per-category and per-benchmark breakdowns, the component ablation, the scaling study, failure cases, cost, and the relation to training-based personalisation. Section numbers are those the manuscript cites.

S1. How Conceptrol was run

Conceptrol setup. We use the authors' `ConceptrolIPAdapterPlusXL` unmodified, on the same SDXL-base-1.0 backbone and the same `ip-adapter-plus_sdxl_vit-h` weights as our method (16 image tokens), at 768×768, 30 DPM-Solver++ steps, CFG 7.5, seed 42. Conceptrol derives its concept masks from text cross-attention and has no layout planner, so it receives the global prompt and the same per-region references; this is its intended usage. We make one change, which favours the baseline. Its helper locates a concept's tokens with `tokens.index(subject_tokens[0])`, i.e. the *first* occurrence, which returns overlapping or misplaced spans on 36 of our 150 prompts—the person and mixed splits routinely repeat a head noun ("a teenage girl … a teenage girl"). We replace it with an ordered subsequence search that advances a cursor past each match, giving disjoint, correctly placed spans on all 150 prompts. Conceptrol therefore receives better concept indices than its own helper produces. We also swept the injection scale over $\{0.8, 1.0\}$ across the full benchmark and report the stronger result.

S2. How MS-Diffusion was run

MS-Diffusion setup. We run the authors' released code and `ms_adapter.bin` on SDXL-base-1.0 with the CLIP-G image encoder they specify, at 768×768, 30 DPM-Solver++ steps, CFG 7.5, seed 42 and our negative prompt, so the two methods differ in mechanism rather than in sampling configuration. Each planned region is passed as one normalised box and the head noun phrase of its sub-prompt as the matching phrase. Two details favour the baseline. Its span helper matches

on whitespace-split words and strips trailing punctuation on the prompt side only, so a phrase carrying an internal comma silently resolves to the empty span $[0, 0]$ and the layout guidance is disabled; we verified on all 150 prompts that the phrases we pass resolve to non-empty, mutually disjoint spans. Adjacent full-height boxes also leave no background region, so we additionally ran the tighter, non-adjacent box shape used in the authors' own examples, and report the stronger of the two. As with the other baselines, the injection scale was swept and the better setting reported.

S3. What the ArcFace averaging convention flatters

ArcFace is averaged over the regions where a face was found, and that flatters the weak rows. Our ArcFace columns follow the usual practice of averaging over regions in which a detector finds a face. A method that renders one subject instead of two is then scored on the one it rendered, and a method that renders neither contributes nothing to the mean—so failing to produce a subject costs nothing. Detection rates differ enough across methods for this to matter. On the 50 person samples (115 regions) they range from 45% to 96%.

Table S1: Reported ArcFace against the same score with missing faces counted as zero. Person samples of Mixed-Eval-v1, 115 regions. The ranking is unchanged, and the margin over every baseline widens, because the baselines are the ones that often render no face at all.

Method	Detected	ArcFace (drop)	ArcFace (zero)
IP-Adapter (global)	45%	0.113	0.051
RPG-only	59%	0.129	0.076
Attention gating	70%	0.118	0.082
IP-FaceID	74%	0.216	0.160
MS-Diffusion	97%	0.290	0.280
Ours	96%	**0.354**	**0.339**

Our own score barely moves ($0.354 \rightarrow 0.339$) because we render a face in 96% of person regions; the text-only baseline loses 41% of its score and global injection loses 55%, since both frequently produce no recognisable face for the detector to score. The advantage over the text-only baseline therefore grows from $2.7\times$ to $4.5\times$ under the stricter policy. We report the drop policy in Table 1 to stay

comparable with prior work and give the zero policy here, rather than choosing the one that suits us.

Global IP injection overwrites the layout. Without regional control the reference's features permeate the canvas, and the Margin is the most negative of any method here: every region crop resembles the *wrong* reference more than its own. Fig. 1 shows the mechanism—conditioned on the first reference, the model renders that subject across the whole frame and the second concept is absent. Prompt adherence and per-region face identity fall with it, and identity specificity falls to 0.076, *below* the 0.082 of a text-only run that never sees a reference at all: the second region's crop now matches a distractor.

Conceptrol writes identity but does not separate the concepts. Run from the authors' code, Conceptrol beats both text-only regional diffusion and global injection on the identity axes: its attention-derived masks do transfer appearance. What it does not do is keep the concepts apart. Its Margin is +0.004—each region crop is as similar to the reference it was *not* assigned as to the one it was—because given a two-subject prompt it typically renders a *single* merged subject carrying attributes of both, rather than two subjects in two places (Supplementary S5). That also explains its low prompt adherence: one merged subject does not satisfy a prompt naming a left and a right one. Our method reaches Margin +0.488 on the same inputs, with DINO-ID 1.9× and ArcFace 1.7× higher.

S4. Why FID and KID rank these methods backwards

Both distributional metrics reward failing the task. We report FID and KID and rely on neither, because the ordering they produce is structural. The reference distribution is the 750 Mixed-Eval-v1 reference photographs, each showing a *single* subject, while every method is asked for *two or three* in one frame. No method that performs the task can match that distribution; the way to score well is to render one clean, centred subject. Global IP-Adapter does exactly that, taking the best FID and KID while having the worst Margin, the lowest CLIP-T, and only one of the two requested concepts present—half the task missing is invisible to both metrics. Conceptrol is second on both for the same reason.

The inversion is systematic rather than anecdotal. Among the five methods in Table 1 that produce well-formed images—global IP-Adapter, Conceptrol, RPG-

only, MS-Diffusion and ours—ranking by concept separation and ranking by distributional distance agree almost perfectly, in the wrong direction: Spearman $\rho = +0.90$ between Margin and FID and +0.80 between Margin and KID (both lower-is-better), i.e. the better a method separates the concepts it was asked for, the further its output distribution moves from the reference photographs. The two methods that actually render every requested subject, MS-Diffusion and ours, are also the two worst on FID (127.4 and 132.6) among the five. We state the restriction explicitly—adding IP-FaceID, which fails the task *and* degrade image quality, collapses the correlation to +0.11 and +0.39—so the claim is that among methods producing comparable image quality these metrics run backwards, not that they do so universally. The wider KID gap is the same statement made with a less biased estimator, not new evidence about our image quality. A second caveat compounds the first: 150 images against 750 references is far from the regime in which FID's Gaussian estimator [1] is stable, which is why the unbiased kernel estimator [2] is reported next to it. What the two do establish is that our outputs are not degenerate—both values stay

S5. Did the Concept Land in the Region It Was Given?

Margin and specificity ask whether a region resembles the right reference. Neither separates two failures that score alike: a region that renders the wrong subject, and a region that renders nothing the prompt asked for. We therefore add a threshold-free check of placement. For each region i we crop it and score the crop with CLIP against every one of that sample's own regional sub-prompts; region i is *bound* when its crop scores highest on sub-prompt i. The comparison is internal to a single image, so no threshold and no cross-sample calibration enters, and chance is exactly $1/N$.

Global injection binds at chance. IP-Adapter applied globally scores 0.418 against a chance level of 0.417: +0.001. On this measure the reference does not merely dominate the layout, it erases it—knowing which region a crop came from tells you nothing about which sub-prompt it satisfies.

Attention gating barely moves placement either. The full-strength gated adapter scores 0.632 against 0.610 for the text-only baseline that never sees a reference—

Table S2: Region-level text-binding accuracy. Fraction of regions whose crop matches its own sub-prompt better than any sibling sub-prompt. Chance is $1/N$: 0.50 at N=2, 0.33 at N=3. "−chance" subtracts the per-sample chance level. Significance is against Ours, paired over the same 150 samples.

Method	N=2	N=3	All	−chance
RPG-only	0.820	0.507	0.663	+0.247
IP-Adapter (global)	0.507	0.320	0.413	−0.003
Conceptrol	0.540	0.382	0.461	+0.044
Attention gating, full strength	0.847	0.502	0.674	+0.258
MS-Diffusion	0.967	0.871	0.919	+0.502
Ours (τ=0.7)	**0.973**	**0.902**	**0.938**	**+0.521**

the same shape of result as Section 3.4, on a metric that has nothing to do with identity. Sub-canvas compositing is +0.271 above it (p<0.001), so the gated injection accounts for about 7% of the placement improvement available from the same adapter.

Against MS-Diffusion this is where we lead, and the lead appears only when a third region is added. Overall we score 0.938 against 0.919, a paired difference of +0.019, 95% CI [+0.012, +0.072], p=0.006. Split by region count the two halves are very different: at N=2 the comparison is a tie (−0.013, p=0.42), and at N=3 it is +0.098, CI [+0.050, +0.145], p=0.0001. Two-region layouts are easy enough that a trained layout-conditioned model places concepts as reliably as sub-canvas denoising does; the separation appears with the third region. Section S11 follows this to N=6 and finds that it does not continue indefinitely.

Taken too far, the release costs placement—but not at the default. Binding accuracy is the only metric in this paper not computed from a fixed region crop, so it is the only one that can see what the global phase does to the boundary the planner drew. It shows a cost that appears only when the release is taken well past the point at which identity is settled: τ=0.7 scores 0.938, indistinguishable from pure sub-canvas compositing at τ=1 (0.928), while τ=0.5 loses 0.026 (p=0.012) and τ=0.3 loses 0.100. The global phase that unifies lighting and background also softens that boundary and occasionally carries a concept across it, and the longer it runs the more often that happens. Our default sits where the effect is not yet measurable; we report the rest of the curve because none of the identity metrics can see any of it.

S6. Cross-Dataset Generalisation (DreamBench-2)

To assess generalisability beyond Mixed-Eval-v1, we evaluate on *DreamBench-2*: a 150-sample two-concept benchmark constructed from DreamBench++ [3] covering 30 concept pairs (HH = 8, HA = 8, HO = 8, AO = 6) with 5 prompts per pair. Table S3 reports DINO-ID, Margin, and CLIP-T, all computed at 1024×1024, the same protocol as every other table.

Table S3: DreamBench-2 results (150 two-concept samples from DreamBench++ [3], 1024×1024). Ours is evaluated here at τ=1, which is the weaker setting on every identity axis (Table 5), so these numbers understate the method relative to its default τ=0.7.

Method	DINO-ID↑	**Margin↑**	CLIP-T↑
RPG-only	0.333	+0.265	**0.656**
IP-Adapter (global)	0.360	−0.054	0.589
MS-Diffusion[¶]	0.705	+0.640	0.639
Ours	**0.806**	**+0.752**	0.623

The comparison against the trained model transfers too. Against MS-Diffusion on this second benchmark we lead by +0.100 DINO-ID and +0.112 Margin and trail by −0.017 CLIP-T, all three at $p<0.001$ over the same 150 samples—the same three signs as on Mixed-Eval-v1, with every magnitude larger. The text-alignment deficit is larger here for the reason the next paragraph gives, and we regard the widening of all three as a property of the benchmark rather than of the method: DreamBench-2's references are photographs of specific subjects, which rewards a method that reproduces the given instance and penalises one that renders the setting the reference came with.

The gains transfer to an unseen benchmark. On DreamBench-2, whose concept pairs and reference photographs are drawn from a different source than Mixed-Eval-v1, sub-canvas compositing again lifts DINO-ID far above both baselines—0.806 versus 0.333 for text-only RPG (2.4×) and 0.360 for global IP-Adapter—and raises the Margin to +0.752, against +0.265 for RPG-only and −0.054 for global injection. The contrast between the two baselines reproduces the Mixed-Eval-v1 pattern exactly: global injection buys appearance fidelity (its DINO-ID exceeds text-only RPG) and pays for it in isolation (its Margin is the only negative one) and

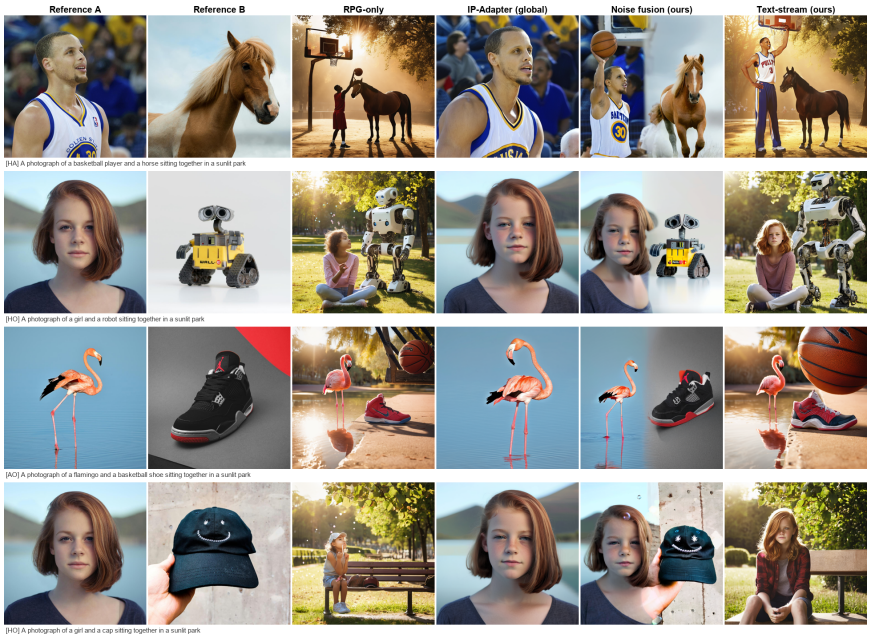


Figure S1: Both operating points on four further DreamBench-2 pairs. Same prompts, same seed, 1024. *RPG-only* satisfies the prompt and ignores the references: the scene is a park, the subjects are generic. *IP-Adapter (global)* reproduces the first reference and drops the second entirely. The two rightmost columns are this paper's two operating points, and the trade between them is visible on every row. *Noise fusion* is the only configuration that reproduces the particular object— this robot rather than a robot, this shoe rather than a shoe—and it pays for that with the panel boundary the two regions meet at. *Text stream* composes one scene with a horse standing in it, a girl seated on the grass, a shoe on the ledge beside the water, and it loses the object's identity in doing so: the robot is no longer WALL-E and the cap is absent. Neither column is uniformly better, which is why we report both rather than designating one as the method.

in prompt adherence (CLIP-T 0.589 versus 0.655). The larger absolute numbers than on Mixed-Eval-v1 reflect the benchmark's simpler two-region layouts; the ordering and the size of the gap over the baselines are consistent across both datasets, indicating the improvement is a property of the method rather than of a particular benchmark.

CLIP-T sits between the two baselines, and the reason is worth stating plainly. Our CLIP-T (0.623) is below text-only RPG (0.656) and above global IP-Adapter (0.589): conditioning on a reference pulls each region away from the generic rendering a text prompt alone produces, and doing so regionally costs less than doing it globally. On this benchmark the effect is larger than on Mixed-Eval-v1, and the cause is visible in the outputs. DreamBench-2 references are photographs carrying their own scenes, our regions tile the whole canvas so none of it is denoised as background, and a region therefore tends to reproduce its reference's

setting along with its subject—so a prompt that relocates the subjects is only partly honoured. This bounds how the DINO-ID column should be read as well: similarity to a reference is raised by reproducing its background, and the gain over the baselines is larger here than on Mixed-Eval-v1 partly for that reason. We return to it in Section 4.6.

S7. Qualitative Analysis

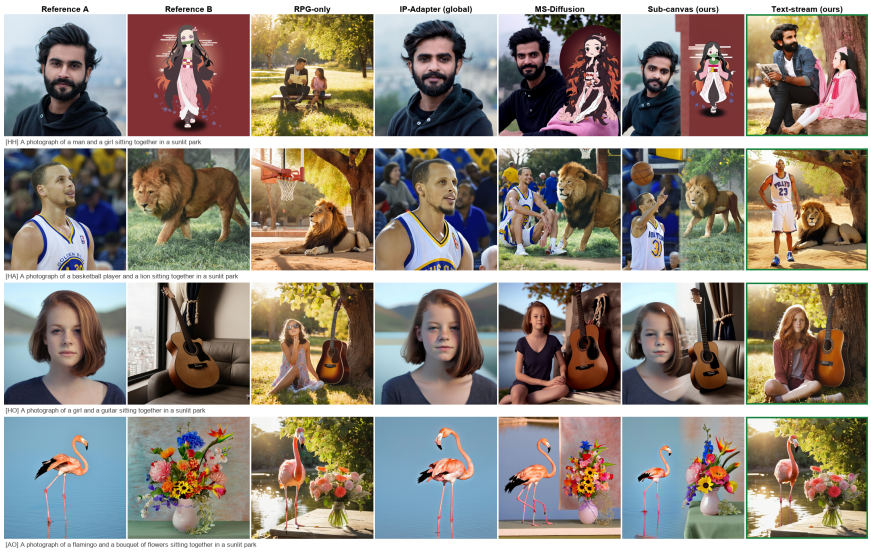


Figure S2: Qualitative comparison on DreamBench-2, whose references are photographs of specific real subjects. One sample per pair type, selected by a fixed rule—the lowest sample index within each type—rather than by inspection. Sub-canvas compositing recovers both references but frames them as two panels meeting at the planner's boundary; the text-stream route puts them in the scene the prompt describes, at the cost of object identity (row 3's guitar and row 4's bouquet are generic, and row 4's fusion is skipped entirely because neither subject is a person). Row 1 shows a further consequence of carrying identity as a face rather than as an image: the stylised reference is rendered as a photograph of a girl, since the fusion transfers facial identity and not the reference's illustration style.

Fig. S2 shows three representative *person* comparisons; Fig. S3 extends the comparison to *object* and *mixed* categories.

Person samples. IP-Adapter (global) renders a single person, faithful to Reference A, filling the frame; the second identity is absent rather than merely contaminated. Conceptrol produces one person too, but carrying traits of both references at once. Our sub-canvas compositing places two distinct people, each matching the reference assigned to its region.

Object and mixed samples. Fig. S3 shows the same pattern outside the face domain. Given a watch and a chair, IP-Adapter (global) returns the watch alone, and Conceptrol returns a single hybrid object that fuses the two. Our method places both objects, each in its planned region, across all category types.

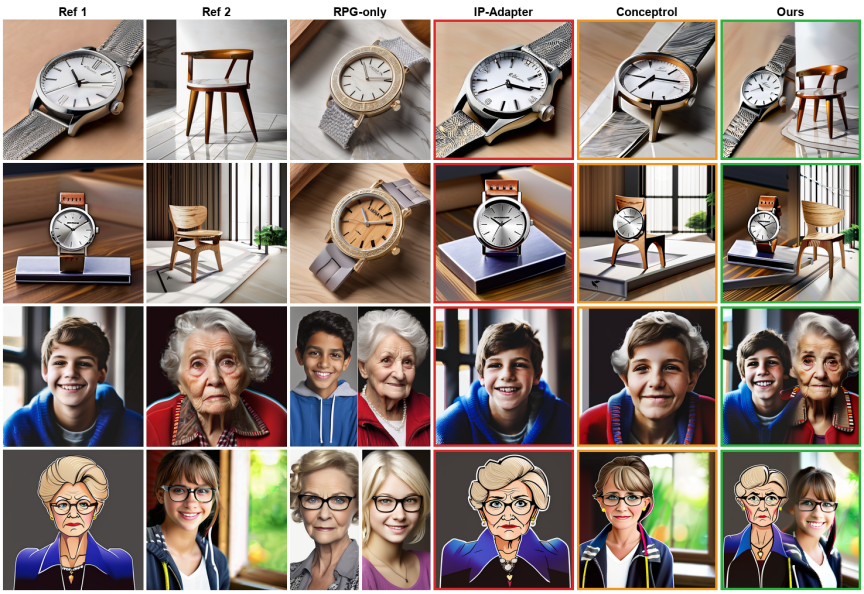


Figure S3: Cross-category generalisation. Top two rows: object samples. Bottom two rows: mixed-scene samples. IP-Adapter (global, red border) leaks one object's texture or colour into the other region. Our method (green border) correctly assigns each reference to its own region across all category types.

S8. Ablation Study

The cascade a reader would propose, run to its end. The rungs above gate the adapter inside a pipeline driven by a single global prompt. The obvious objection is that we never combined the two halves that matter: RPG's regional *text* attention, which is what makes its compositions read as photographs, and Conceptrol's semantic mask, which is what keeps a reference's texture off the background. Cascade them—the planner supplies M_{spatial}, the region's own text attention supplies M_{semantic}, and the adapter writes only where both agree—and the identity should arrive without the leakage.

We implemented that cascade inside RPG. The semantic map costs nothing extra there: RPG's regional attention already computes the queries and keys the

Table S4: Where the spatial constraint is applied. All rows use the same MLLM plan, the same references and the same adapter weights; they differ in *where* the regions are separated and, for the two attention-gated rows, in how strongly the injection is attenuated. *Attenuated* stacks the per-layer, per-timestep and per-category gates that a leakage-suppressing configuration typically applies; *full strength* removes all of them and runs at the same λ=0.8 as our method. Neither exceeds the text-only baseline on any identity axis. Moving the constraint into the sampling loop does, and denoising per-region sub-canvases rather than the full canvas is what additionally fixes the chimeric-subject failure. Binding accuracy (Section S5) separates the same rows independently of identity: 0.632 for attention gating against 0.610 for the text-only baseline, and 0.938 for ours. All rows use τ=1 (late release off) so that this table isolates the fusion geometry; the default τ=0.7 used elsewhere is evaluated separately in Table 5.

Separation mechanism	DINO-ID	Margin	CLIP-T	ArcFace	Spec.
None (text only, RPG)	0.352	+0.076	0.650	0.129	0.082
None (global adapter)	0.341	−0.082	0.615	0.089	0.076
Attention, box mask	0.358	+0.098	**0.652**	0.122	0.089
Attention, semantic mask	0.387	+0.004	0.630	0.165	0.090
Attention, box ∩ semantic	0.593	+0.353	0.651	0.243	0.189
Noise fusion, full-canvas	0.603	+0.398	0.641	0.212	0.185
Noise fusion, sub-canvas (Ours)	**0.719**	**+0.490**	0.646	**0.264**	**0.249**

map is read from, so no hook and no second forward pass are needed. We then ran it at full injection strength on 24 photographic human-human pairs, framed as medium shots so that faces occupy about 9.5% of the canvas and identity has room to be visible.

Table S5: The RPG cascade delivers RPG's pictures and none of the identity. ArcFace on detected faces, so the score does not move with framing; 0 is chance. Text-only RPG is the floor. Adding the adapter, with or without the semantic mask, and at five times the default strength, never leaves it.

Configuration	ArcFace	Faces/img	Face area
RPG, text only	+0.014	1.96	9.48%
+ IP-Adapter, box mask	+0.003	1.88	4.97%
+ IP-Adapter, box ∩ semantic	−0.010	1.79	7.31%
the same, binarised mask	+0.004	1.88	5.21%
box ∩ semantic, λ=1.5	+0.014	1.88	6.20%
box ∩ semantic, λ=3.0	−0.008	1.75	4.54%

Table S5 is the answer, and it is flat. The pictures are RPG's—medium framing, natural poses, no boundary—and the people in them are not the reference people. Raising λ from 0.6 to 3.0 does not change that; it only degrades the image, blurring faces and oversaturating colour while the face count falls. The semantic mask cannot help, because a mask restricts where identity is written and there is no

identity to restrict: the box-gated row is already at chance.

This closes the question the ladder opens. Attention-level gating recovers 1.7% of the identity the same adapter delivers from the sampling loop when measured in the Conceptrol pipeline; measured inside RPG, with the strongest mask the two systems can jointly produce and five times the injection strength, it recovers none. The sub-canvas construction is not a preference over a simpler cascade. The simpler cascade does not work.

The site of the constraint, not its presence, determines identity. The two attention-gated rows bracket the injection strength available to that formulation—one heavily attenuated, one with every gate removed and λ raised to our own setting—and neither approaches the rest of the table. The attenuated variant reproduces the text-only baseline literally, on 41 of 150 images bit-for-bit. The full-strength variant reproduces none of them and does clear the baseline on DINO-ID and Margin, by +0.018 and +0.022 respectively, but leaves ArcFace and specificity (Section 4.3) untouched and closes only about a twentieth of the distance to noise-level fusion. Injection strength is therefore not what separates these rows from the rest of the table. Relocating the identical adapter to noise-level fusion lifts DINO-ID to 0.561 and Margin to +0.350 without any change to the adapter or its scale—the gain comes from *where* the spatial constraint is imposed.

We note that recent training-free multi-concept methods [4, 5, 6, 7, 8] differ in how they compute or refine the attention mask but all inject and confine identity *within* the attention operator, i.e. they instantiate Eq. 3. Our attention-gated rows are one implementation of that formulation, not a reimplementation of any of these methods, and we do not claim to have measured them: a different masking heuristic could plausibly behave differently. What our measurement does establish is that the shortfall is not explained by weak injection, since removing every attenuation does not move any identity axis.

Sub-canvas vs. full-canvas. Full-canvas fusion already recovers most of the identity, but each per-concept pass composes its subject at the centre of the frame, so for centred subjects the masks splice two half-subjects into one. Confining each pass to its own sub-canvas raises DINO-ID further ($0.561 \rightarrow 0.647$) and Margin ($+0.350 \rightarrow +0.415$), and—more importantly—removes the chimeric-subject artefact that the metrics only partly capture (Fig. S4). ArcFace is unchanged (0.268 vs. 0.269)

because it is measured on the face crop, which is intact in both; the difference is whether that face belongs to one coherent person or is stitched from two.

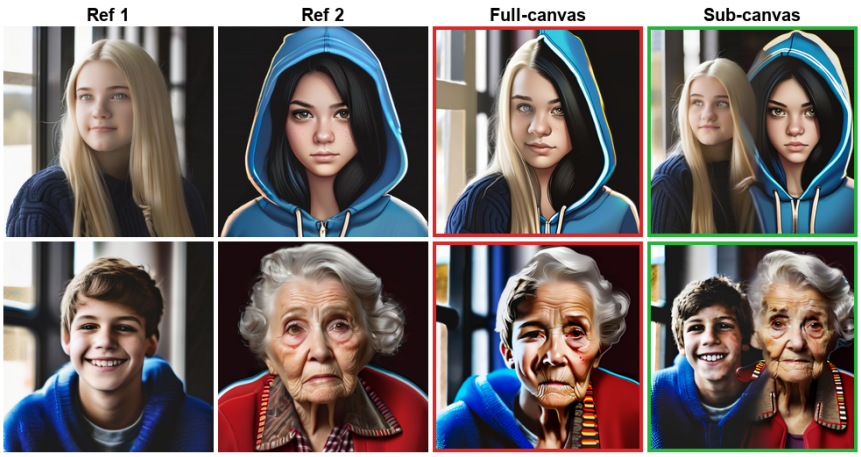


Figure S4: Full-canvas vs. sub-canvas fusion. Same inputs, same full-strength adapter, same fusion weights. *Full-canvas*: each concept is denoised over the whole frame, so both subjects form at the centre and the masks merge them into a single chimeric face. *Sub-canvas*: each concept is denoised inside its own region, so the two subjects are distinct and correctly placed. Both columns use τ=1 to isolate the fusion geometry from the late release of Section 4.5.

S9. Where the Release Point Comes From

Table 5 is an observational sweep, and a flat curve admits two readings. Identity may be settled before the release, so the late steps *cannot* move it; or the late steps may retain the ability to move it and merely not exercise it on this benchmark. The design of Section 3.6 assumes the first, as does the wider literature—Personalize Anything [9] injects aggressively early and regularises late on that premise, and acceleration work reports identity stabilising within the first handful of steps [10]—but a sweep cannot distinguish the two. An intervention can.

Protocol. We denoise each of the 30 reference-swap samples of Section 4.4 with region 0 carrying its own reference A for the first k steps and the same-category alternative A' for the remaining $30 - k$. The regional decomposition runs for the whole trajectory (τ=1), so the reference is the only thing that changes at step k; prompt, layout, seed and region 1's reference are untouched. On region 0's crop

we report

$$A(k) = \mathrm{sim}(\mathrm{crop}_0, A') - \mathrm{sim}(\mathrm{crop}_0, A), \tag{S1}$$

$$F(k) = D(k) \,/\, D(0), \tag{S2}$$

with D as in Eq. 9, measured against the never-swapped render. $A(k)$ asks which reference the output ended up resembling and needs no anchor; its zero crossing is the commitment point. $F(k)$ is the fraction of the achievable identity change that a swap at step k can still buy. We also report $S(k)$, the pixel distance from the never-swapped render, to separate *the trajectory stopped responding* from *the trajectory still responds but no longer with identity*.

k=0 and k=30 are boundary controls rather than measurements. A further control is available because k=30 is, by construction, an ordinary τ=1 render: it reproduces the independently generated τ=1 image of the same sample to a mean absolute difference of 0.03 of 255, so the probe samples the trajectory this paper reports and not a re-implementation of it.

Table S6: Identity commitment. Region 0's reference is replaced by a same-category instance at step k of 30, with the regional decomposition active throughout and everything else fixed (30 samples). S: pixel distance from the never-swapped render. A (Eq. S1): positive when the output resembles the newly supplied reference, negative when the first k steps had already decided. F (Eq. S2): fraction of the achievable identity change still available at step k. p is a two-sided paired t test on A. The choice of subject stops being reversible at $k\approx6$, a fifth of the way through the trajectory.

k	$k/30$	S	A	p	F
0	0.00	55.75	+0.481	< 0.001	1.000
2	0.07	49.77	+0.405	< 0.001	0.916
4	0.13	41.71	+0.258	< 0.001	0.751
6	0.20	32.46	+0.034	0.61	0.538
9	0.30	21.27	−0.222	< 0.001	0.323
12	0.40	11.96	−0.368	< 0.001	0.188
15	0.50	6.11	−0.456	< 0.001	0.128
30	1.00	0.00	−0.590	< 0.001	0.000

The choice of subject is made in the first fifth of the trajectory. $A(k)$ crosses zero at k=6.2 and $F(k)$ falls through a half at k=5.9: two estimators, one anchored and

one not, agree on $\tau^* \approx 0.21$. Before step 6 the incoming reference still wins; after it, the output resembles the reference the first six steps saw, however many steps the new one is given. By k=15—half the trajectory, and well past our default release point—a swap buys only 13% of the achievable change and the output still sits 0.397 closer to the original reference than to the one actually driving 24 of the 30 steps. One caveat bounds this: the sub-prompt is written for A and is not swapped, so A carries a standing text advantage and the crossing is a lower bound on where the reference alone could still reverse the choice.

The trajectory keeps responding after it stops carrying identity. At k=6 the swap still moves the image by S=34.7 of 255—three fifths of a full-trajectory swap—while A is indistinguishable from zero. This is the dissociation of Section 4.4 in time rather than across methods: late injection perturbs the image without changing whose identity it carries. A study that measured only how much the output moved would place the commitment point far later than it is.

The release point is not set by identity. If identity commitment were the constraint, τ=0.21 would suffice. It does not: at τ=0.3 binding accuracy is 0.828 against 0.928 at τ=1, still a tenth short, and it does not saturate until $\tau \approx 0.7$ (Table 5). What the regional phase buys between τ^* and our default is therefore not the choice of subject, which was settled at 0.21, but the isolation of the subjects from each other. The trade-off the switch point actually governs is isolation against whatever the released phase provides—the same conclusion Section S5 reached from the opposite direction, that the release costs placement rather than identity—and we set τ at 0.7, where isolation is still at its maximum and the identity settled at 0.21 is not disturbed.

S10. Per-Category Analysis

Identity gains are strongest where faces are present. Table S7 breaks ArcFace down by category. The detection rate is as informative as the score here. Global IP injection reaches ArcFace 0.103 on person samples—close to text-only (0.130)—but its detection rate collapses from 0.76 to 0.55, because the frame that should hold two faces holds one; the score is computed on whichever single face survives. Sub-canvas compositing instead lifts ArcFace by roughly 80% on person (0.130 →

Table S7: Per-category ArcFace identity on Mixed-Eval-v1 (Ours at the default τ=0.7). Person: portrait samples; Mixed: scene samples with faces. Detect rate shows fraction of regions where ArcFace detected a face.

Method	Person		Mixed		Overall
	ArcFace↑	Det.Rate	ArcFace↑	Det.Rate	ArcFace↑
RPG-only	0.130	0.76	0.133	0.56	0.131
IP-Adapter (global)	0.103	0.55	0.093	0.55	0.101
Ours	**0.265**	0.88	**0.282**	0.58	**0.272**

0.235) and mixed (0.133 → 0.271) splits *and* raises the person detection rate to 0.88—the generated faces are more numerous, better formed, and closer to the reference identity at the same time.

S11. Scaling to More Regions

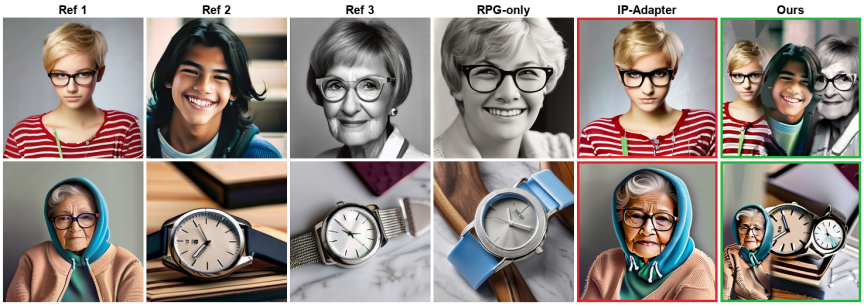


Figure S5: Three-region generalisation (N=3)**.** The construction extends to three simultaneous reference concepts without modification. IP-Adapter (global) is conditioned on one reference and renders it alone, so the other two concepts are absent. Sub-canvas compositing places all three, and the quantitative form of that observation is the N=3 column of Table S2 (0.898 against 0.800, p=0.0001). The second row also shows the failure the geometry does not prevent: with one person and two similar objects the person is composed inside a region narrower than a portrait wants, and is rendered at a scale that does not match the objects beside it. Region width is fixed by the planner's split ratio and is not adapted to what each region contains (Section 4.6).

The construction imposes no upper bound on N: the MLLM planner assigns one reference per region, and the N sub-canvases $\{\Omega_i\}$ partition the latent, each denoised by its own single-concept adapter pass and fused by Eq. 5. Fig. S5 shows N=3 person samples from Mixed-Eval-v1: the per-region isolation holds, whereas global injection produces an indistinct blend across all three regions.

Whether it keeps holding is a measurement rather than a corollary, and the structural argument runs in our favour: a sub-canvas is denoised in its own window,

so a region's problem does not depend on how many other regions exist, whereas a trained layout-conditioned model has no such guarantee. We tested it. *Scaling-Bench* extends the same task to $N \in \{4, 5, 6\}$ with 30 samples per region count (90 total, 10 per category per count), scored with the region-level binding accuracy of Section S5. That metric is the one instrument here that stays interpretable as N grows, since it is internal to a single image and its chance level $1/N$ is known exactly.

Table S8: Region-level binding accuracy against region count. Scaling-Bench, 30 samples per N; the N=2, 3 columns are the Mixed-Eval-v1 numbers of Table S2, repeated for continuity. Δ and p compare our default (τ=0.7) against MS-Diffusion, paired and two-sided; the τ=0.5 row is kept to separate the release point's contribution from the method's. Both methods stay above chance at every N, and the gap between them grows monotonically with the number of regions, from +0.006 at two to +0.194 at six.

N	2	3	4	5	6
chance	0.500	0.333	0.250	0.200	0.167
MS-Diffusion	0.967	0.871	0.725	0.600	0.439
Ours (τ=0.5)	—	—	0.808	0.700	0.539
Ours (τ=0.7)	**0.973**	**0.902**	**0.825**	**0.713**	**0.633**
Δ (τ=0.7)	+0.006	+0.031	+0.100	+0.113	+0.194
p	0.66	0.18	0.060	0.020	<0.001

The structural argument survives, and it is the clearest result in the paper's favour. The two methods do not degrade at the same rate: least squares over N=2 . . . 6 gives −0.133 accuracy per added region for MS-Diffusion against −0.087 for ours, a third slower. The gap grows monotonically—+0.006, +0.031, +0.100, +0.113, +0.194—and becomes significant where it matters most, at N=5 (p=0.020) and N=6 (p<0.001), the region counts at which a trained layout-conditioned model is closest to chance. At six regions MS-Diffusion retains +0.272 above chance and we retain +0.466.

This is what the sub-canvas construction was supposed to buy and, at 768, did not. We report the earlier result because the reason for the difference is instructive. A sub-canvas is denoised at the size of its own region, so at N=6 on a 768 canvas each pass runs on a 128-pixel-wide strip, far outside the range SDXL was trained on; at 1024 the same strip is 171 pixels. What limits the construction at high N is therefore not the per-region independence but the resolution that independence

leaves each region, and a backbone evaluated below its native size hides exactly the advantage the construction is meant to have. A trained model that reasons over the whole canvas has no such floor, which is why it loses less from the smaller canvas and gains less from the larger one.

The advantage is in placement, not in usable images. Fig. S6 puts the three region counts beside each other, and it qualifies the result above in a way the table cannot. At $N \geq 4$ neither method produces an image that would survive any practical use: ours breaks into strips with visible stylistic seams, MS-Diffusion holds a more photographic composition while placing fewer concepts where they belong. What Scaling-Bench establishes is therefore that the sub-canvas construction keeps assigning concepts to their regions further into a regime where both methods have already stopped producing usable output—a statement about the mechanism's degradation, not a claim that the method is useful at six regions. The practical range of the method, on this evidence, is the $N \leq 3$ of Mixed-Eval-v1.

The release point contributes little of this. Holding resolution fixed, moving τ from 0.5 to 0.7 is worth +0.017 at N=4 (p=0.48), +0.013 at N=5 (p=0.70) and +0.094 at N=6 (p=0.001). Only the last is significant and we report it uncorrected for three tests, so the honest reading is that the release point costs little at any region count rather than that it helps at high ones. The separation above is the geometry's, not the schedule's.

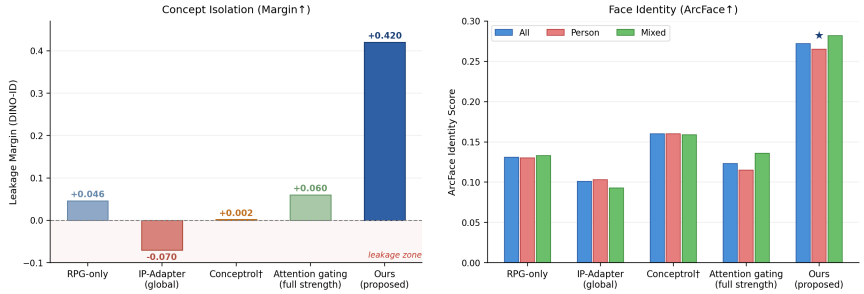


Figure S7: Metric comparison. Left: Leakage Margin. Global IP-Adapter falls below zero (leakage zone, red shading); Conceptrol† and attention gating at full strength sit at essentially zero, i.e. each region crop matches the wrong reference as well as its own; only sub-canvas compositing separates the concepts. Right: ArcFace identity by category. Conceptrol transfers more face identity than the other baselines but, as the Margin shows, into a single merged subject rather than two.



Figure S6: What the scaling advantage does and does not look like. One sample per region count, chosen by a fixed rule: within each count we show the sample whose binding-accuracy gap is closest to the mean gap for that count, so the row is typical of the measured difference rather than its best case. The measured advantage is real—our regions hold the concepts they were assigned more often—but the figure shows what the number does not: at $N \geq 4$ neither method produces an image a user would keep. Ours fragments into strips with visible stylistic discontinuities; MS-Diffusion stays more photographic while binding fewer regions correctly. Part of the disorder is the benchmark's, since Scaling-Bench draws its references from the Mixed-Eval-v1 pool whose styles are uncontrolled (Section 4), and part is the method's. We report the figure beside the table because the table alone would overstate the result.

S12. Failure Case Analysis

Despite strong average performance, our method exhibits recurring failure modes that reveal the limits of training-free spatial conditioning. We categorise the worst-performing samples (DINO-ID < 0.16) from Mixed-Eval-v1 into four patterns:

(F1) Crowded three-region layouts. When N=3 equal-width regions are used, each strip is only ∼341 px wide at 1024 px resolution. The IP-Adapter cross-attention guidance spreads across the narrow strip and cannot fully suppress leakage from adjacent regions. Sample `mixed_3r_039` (mug | teenager | plant) achieves DINO-ID = 0.107 and Margin = −0.133—the worst leakage in the dataset.

(F2) Visually generic objects. Everyday objects such as apples, mugs, and books produce nearly identical DINOv2 embeddings across different instances. Even when the generated image renders the correct object category, the metric registers low identity because DINOv2 encodes instance-level texture rather than category membership. Samples `object_3r_032` and `object_3r_047` (apple/watch/mug and apple/book/plant, respectively) illustrate this: the model generates plausible objects, but the embedding comparison yields ID < 0.16.

(F3) Contradictory textual descriptions. Some sub-prompts assign conflicting attributes to one subject—"long black hair" and "short blonde hair" in `person_3r_029`, for instance—and both the text encoder and the adapter respond with a blended or incoherent appearance (DINO-ID = 0.150). The fault is ours rather than the planner's: it originates in Mixed-Eval-v1's prompt construction, which draws two attributes without excluding pairs from the same group, and affects 11.2% of person sub-prompts (Section 4). The planner propagates the contradiction faithfully. We keep these samples in every table because removing them after the fact would be selection on the outcome, and because the contradiction is identical for every method compared.

(F4) Face identity in narrow person strips. Three-person layouts force each face into a ∼341 px strip. ArcFace detection success drops below 50% in these cases, and even when faces are detected, the limited spatial extent weakens the IP-Adapter conditioning signal relative to the background. `person_3r_049` achieves DINO-ID = 0.150 and Margin = −0.077.

These failure modes suggest that future work should focus on adaptive region

sizing guided by concept salience, and reference denoising strategies that increase robustness to view-angle variation.

Single-concept identity methods do not extend to this setting. We ran InstantID [11] at 768 before settling on the native-resolution protocol and do not carry the row forward, but the failure is worth one sentence because it is structural rather than incidental: the method derives face keypoints from *one* reference and applies them as a ControlNet condition to the *entire* canvas, so in a two-person layout a single face geometry is imposed over both regions and neither crop passes ArcFace detection on any of the 150 samples.

S13. Computational Efficiency

Table S9 reports per-image denoising time and peak GPU memory on a single NVIDIA RTX 5080 (16 GB) at 768×768, 30 DPM-Solver++ steps. Every configuration is measured with the *identical* inline sampler, no CPU offloading, and the VAE decode excluded, so the numbers are directly comparable and reflect the denoising loop alone. A single full-canvas conditioned pass—the per-step cost of global IP-Adapter—is the natural unit of comparison; region-planning baselines such as RPG-only and Conceptrol perform at least one such pass plus their own regional or masking overhead.

Table S9: Inference cost (RTX 5080 16 GB, 768×768, 30 DPM-Solver++ steps, seed 42; `time.perf_counter`, 5 timed runs after 2 warm-ups, identical inline sampler, no CPU offload, VAE decode excluded).

Method	s / image	× IP-global
IP-Adapter (global)	**9.46**	1.00×
Ours	9.88	1.04×
MS-Diffusion	11.92	1.26×
RPG-only	15.64	1.65×
Attention gating	16.00	1.69×
Conceptrol	17.22	1.82×

Sub-canvas compositing evaluates the UNet once per region per step, but each

pass runs on a sub-window narrower than the full canvas, so cost grows *sublinearly* in the number of regions N: two regions cost 1.4× a single full-canvas pass rather than 2×, and three cost 2.0× rather than 3×. Peak memory stays flat at 9.5 GB across N—only one sub-canvas is resident at a time and the footprint is dominated by the fixed SDXL and adapter weights—so the method adds concepts within the 16 GB budget without the memory blow-up of batching N full-resolution images.

S14. Comparison with Training-Based Methods

Our method targets the *zero-shot* setting: given only inference-time reference images, it produces personalised multi-concept images without any per-concept optimisation. Training-based methods—DreamBooth [12], Custom Diffusion [13], and Mix-of-Show [14]—generally achieve higher fine-grained identity fidelity by embedding specific concepts into model weights. However, they require per-concept (or per-concept-pair) fine-tuning before inference, which precludes use when reference images arrive at test time. Table S10 summarises the practical trade-off.

Table S10: Training cost vs. capability comparison. "Inference-time reference" = method accepts unseen reference images without retraining. *FreeCustom is tuning-free but additionally requires an external segmentation model to obtain per-concept masks.

Method	Training	Per-concept time	Multi-concept	Inference-time ref	Backbone
DreamBooth [12]	Full UNet	~30 min	×	×	SD
Custom Diffusion [13]	Key layers	~30 min	Partial	×	SD
Mix-of-Show [14]	LoRA / concept	~20 min	✓	×	SD
MC^2 [5]	Per-concept models	~20 min	✓	×	SD
LaTexBlend [6]	Per-concept bases	~20 min	✓	×	SD
LoRAShop [15]	LoRA / concept	~20 min	✓	×	Flux
FreeCustom [7]	None	—	✓	✓	SD 1.5*
MS-Diffusion [16]	Corpus, once	—	✓	✓	SDXL
FastComposer [17]	Corpus, once	—	✓	✓	SD
XVerse [18]	Corpus, once	—	✓	✓	Flux
IP-Adapter (global)	None	—	✓	✓	SDXL
Ours	**None**	—	✓	✓	SDXL

The distinction that matters is not training in general but *per-concept* training. Methods in the first block bake each concept into weights before inference, so a reference photograph first seen at test time cannot be used at all; quantitative comparison against them would confound the mechanism under study with a strictly easier problem setting, and we do not attempt it. Methods in the second block are trained once on a subject-annotated corpus and are then zero-shot exactly

as we are. Those *are* directly comparable, and the strongest of them that targets our layout-conditioned setting on our backbone, MS-Diffusion, is evaluated head-to-head in Table 1 under a matched protocol. On that benchmark we lead it on identity, placement and isolation and trail it on text alignment (Section 4.2), and we present both halves, including the layout convention under which the isolation result reverses. What our method offers besides is the absence of a subject-annotated training corpus and a lower cost per image (9.88 s against 11.92 s); what the comparison establishes for the argument of this paper is that the separation both methods achieve is unreachable by the training-free attention-gated route, whether it is bought with supervision or with a change of injection site.

On recent multi-concept methods. The 2025 multi-concept literature is dominated by two families, neither of which yields a like-for-like baseline in our setting. MC^2 [5] and LaTexBlend [6] *compose pre-trained single-concept models*: every concept must first be embedded into weights by a per-concept optimisation run, so they inherit the training-based limitation above and cannot consume a reference photograph that first appears at test time. FreeCustom [7] is genuinely tuning-free, but couples a multi-reference self-attention mechanism to an SD 1.5 backbone and additionally requires an external segmentation network to produce the per-concept masks it filters with; differences in backbone capacity and mask supervision would dominate any direct metric comparison against our SDXL pipeline. We therefore compare against the mechanism these methods share—confining identity *inside* the attention operator—rather than against their individual implementations, which is exactly the attention-gating ablation of Table S4. That ablation is backbone-, planner- and seed-matched to our method, and so isolates the design choice under study instead of conflating it with backbone differences.

S15. Where the panel boundary comes from, and what closes it

The control we should have run first, and the conclusion it overturns. Nine failures spread across geometry, receptive field, and post-processing invite the conclusion that the panel structure is intrinsic to noise-level fusion. An earlier draft of this analysis drew exactly that conclusion. It is wrong, and the control that refutes it is the cheapest one available. We ran the method with the same regions, the same overlap, the same schedule and the same fusion rule, changing one thing: the

adapter scale is set to zero, so each sub-canvas is driven by its text prompt alone. The boundary step is 1.059, against 1.103 for the method as reported and 1.37 for a collage, on the same 30 samples. Sub-canvas fusion applied to text-conditioned regions leaves no visible seam. What leaves one is the references: two photographs taken in different places, transferred whole by the adapter—background included—and each reproduced faithfully by the pass that receives it. The boundary is where two reproduced scenes meet.

This reattribution predicts its own repair, and the prediction holds. Segmenting each subject and compositing it onto a common ground before encoding gives the two passes nothing left to disagree about, and the step returns to the text-only floor: 1.062 on flat grey, 1.017 on a shared defocused backdrop. The dose behaves as the account requires—blurring that shared backdrop by 24 pixels rather than 48 leaves enough of the original scene legible for the two passes to keep disagreeing, and the step stays at 1.096, statistically indistinguishable from the method as reported. The price is a small but real loss of identity, −0.035 and −0.033 DINO-ID respectively (both $p<0.01$), because the discarded background was carrying part of what DINO was matching on.

Most of that identity loss is recoverable, and the way it is recovered confirms the mechanism a second time. A binary silhouette composites the subject onto the ground as a cut-out, and the adapter transfers the cut-out edge along with the subject; blurring the segmentation alpha by 24 pixels so that the subject fades into the ground instead recovers two fifths of the loss, to −0.019 DINO-ID and −0.027 Margin, while holding the step at 1.016. Against the unfeathered variant the identity gain is +0.014 ($p=0.046$). The best repair we have is therefore a preprocessing step on the references alone: segment, composite onto one defocused ground, feather the seam. It touches neither the fusion rule nor the layout, which is the point.

When it works and when it cannot. The repair is bounded by what segmentation can extract. Where each reference contains a subject with a silhouette—a person, an animal, an object—the two grounds agree and the boundary closes. Where a reference *is* a scene, as with our stained-glass and multi-panel character-sheet references, there is no subject to lift off it, the segmenter returns most of the frame, and the panel survives every variant above. That is a limitation of the diagnosis and also a small confirmation of it: the boundary persists in exactly the cases where

the two references still carry incompatible contexts after the repair.

Two honest qualifications. The 1.017 figure sits *below* the text-only floor, and it should: with the same blurred image behind both subjects the low-frequency field is continuous by construction, so on this one variant the measure is partly reporting what we built rather than what was generated. And a defocused ground is not the prompted scene—the seam is gone, but so is the setting, which is the same trade Section 3.4 reaches from the other direction. We therefore report the segmented-reference variants as a diagnosis that localises the cause and as an optional preprocessing step with a measured price, not as a replacement for the method whose numbers this paper reports: every comparison in Sections 4.2–S11 is run without it, so that what is compared across methods is the generation procedure and not our choice of segmenter. The two directions that would keep the scene are outside this paper's setting: token replacement in a backbone whose tokens carry semantics, which requires a DiT, and a trained grounding module, which requires the corpus whose absence is the point of a training-free comparison.

S16. The frontier is the backbone's, not the tuning's

Every configuration in this study trades identity against framing, and the trade is remarkably stable: across four mechanisms and roughly twenty configurations, more identity arrives only with a larger face, a tighter crop and a boundary between the regions. Nothing we tried escaped it. A reader is entitled to suspect that a study which fails twenty times has simply not found the right setting.

Concurrent work names the reason, and it is architectural rather than a matter of search. Feng et al. [9] observe that a diffusion transformer represents content in tokens whose position is supplied separately, so a reference's tokens can be injected *without* their positional information, whereas a U-Net's convolutions entangle content with position and reproduce the reference's spatial arrangement along with its appearance. Our measurements are the U-Net side of exactly that statement. A CLIP image embedding fed to IP-Adapter cannot deliver "who this is" without also delivering "framed like this": raising the adapter's strength collapses the prompt's medium shot to the reference's bust (9.5% $\rightarrow$ 20.5% face area, Table 7), and lowering it removes the identity instead. The same entanglement explains why sub-canvas compositing scores highest on identity while sitting

closest to a literal paste, and why attention gating inside RPG recovers no identity at any strength (Table S5): a spatial mask can restrict where entangled content lands, but cannot separate what is entangled.

Read that way, the results here and the DiT results are consistent rather than competing. Token replacement is the rung our ladder ends on and could not climb—on SDXL it improves every metric we report while reproducing the reference almost exactly (Section 4.4)—and on a DiT the same operation succeeds, with early injection for consistency and late regularisation for flexibility [9]. The regularisation is aimed at the failure our copy-fidelity check measures. Work that removes the pose entanglement without changing backbone does so by training for it: Qian et al. [19] disentangle pose and illumination across subjects with a cross-reference objective, which is not available to a training-free method.

We therefore state the scope of this paper's negative results precisely. They are not evidence that regional personalisation is hard; they are measurements of what a U-Net's cross-attention can and cannot be asked to do, with the size of each cost attached. The text-stream route of Section 4.7 is the one configuration that sidesteps the entanglement without leaving the backbone, and it does so by not using the image branch at all—which is also why its scope is limited to subjects whose identity encoder was trained on faces.

S17. Protocol for the baselines in Table 1

†: Conceptrol [4], run from the authors' official implementation (`Conceptrol IPAdapterPlusXL`) with the same SDXL backbone, IP-Adapter Plus ViT-H weights, resolution, scheduler, steps, CFG and seed as ours. Injection scales 1.0 (the authors' default) and 0.8 (matched to our λ) were both run over all 150 samples and the stronger of the two (0.8) is reported; at 1.0 the row reads 0.357/+0.004/0.621/0.155/0.153/130.8/26.8. Conceptrol has no layout planner, so it receives the global prompt, and we supply corrected per-concept token spans (Section S1). ¶: MS-Diffusion [16], run from the authors' released code and `ms_adapter.bin` on the same SDXL-base-1.0 backbone, at the same resolution, scheduler, steps, CFG and seed as ours, and driven by our planner's regions. Four configurations were run over the full benchmark—injection scale {0.6 (the authors' default), 0.8 (matched to our λ)} crossed with box shape

{our adjacent full-height strips, the tighter non-adjacent shape used in the authors' examples}—and the strongest is reported (scale 0.8, tight boxes). The box shape matters more than the scale: with our strips the same scale reads 0.509/+0.327/0.645/0.215/0.212/128.9/23.8, so passing the layout in the authors' own convention is worth +0.112 Margin to the baseline. Ours = the noise-fusion operating point: sub-canvas regional compositing with per-region full-strength IP-Adapter and late global release at the default τ=0.7 (Section 4.5). ‡: IP-Adapter FaceID PlusV2 [20], conditioned globally on the first reference of each sample—its ArcFace identity embedding together with the CLIP ViT-H features—at the same resolution, scheduler, steps, CFG and seed as ours, at the stronger of the two injection scales we ran (0.6). In 81 of the 150 samples the guiding reference contains no detectable face, since the object category is not faces, so the identity embedding degenerates to a zero vector and only the CLIP branch remains active; its DINO-ID lands *below* the text-only baseline as a result. This is inherent to applying a face-specific method to a benchmark that contains objects, and we report it rather than dropping the row. InstantID, which appeared in an earlier 768 version of this table, is dropped rather than carried over at a resolution it was not re-run at; it scored far below every row above on every axis.

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